

Seeing–Saying–Reasoning: A Review of Multimodal Artificial Intelligence in Medical Imaging

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Abstract—The integration of artificial intelligence into medical imaging has advanced rapidly, progressing from early perception-focused systems to multimodal architectures capable of generating clinical language and supporting diagnostic reasoning. Despite this progress, existing surveys organise the literature primarily by model architecture or task type, offering limited insight into the progression of clinical capability that modern systems exhibit. This paper introduces the Seeing–Saying–Reasoning (SSR) framework, which organises medical imaging AI systems according to three functional stages: visual perception of medical images (Seeing), translation of visual findings into clinically meaningful language (Saying), and integration of perceptual and linguistic outputs with medical knowledge to support diagnostic inference (Reasoning). Formally defined as a progression of functional mappings, the SSR framework mirrors the cognitive workflow of clinical interpretation and provides a principled basis for analysing both the capabilities and limitations of existing systems. Using this framework, we present a structured taxonomy of representative multimodal AI systems, analyse the reliability and safety challenges that arise specifically at each stage—including hallucination, weak visual grounding, three-dimensional context loss, and dataset bias—and examine the clinical translation and regulatory considerations that govern trustworthy deployment. We further outline a research roadmap linking open challenges to specific SSR stages. The SSR framework and accompanying analysis provide a capability-oriented perspective on the evolution of medical imaging AI that complements existing architectural surveys and highlights the most consequential gaps on the path toward clinically reliable multimodal intelligence.

I. INTRODUCTION

Medical imaging plays a central role in modern clinical care. Radiological examinations—including computed tomography, magnetic resonance imaging, and chest radiography—are among the most frequently performed diagnostic procedures worldwide, and their interpretation demands significant clinical expertise, sustained attention, and substantial time investment. The growing volume of imaging studies, combined with global shortages of radiologists and pathologists in many healthcare systems, has created strong institutional motivation to develop computational tools capable of supporting or augmenting clinical image interpretation. Artificial intelligence, and deep learning in particular, has emerged as the primary technological response to this challenge over the past decade.

Early progress in applying AI to medical imaging focused predominantly on perception tasks—training convolutional neural networks to detect, classify, and segment pathological findings in specific imaging modalities. These systems

demonstrated remarkable benchmark performance, in some cases matching or exceeding specialist clinicians on narrowly defined tasks. However, their clinical impact has remained limited by fundamental constraints: task specificity, poor generalisation across institutions and patient populations, and an inability to communicate findings in natural language or incorporate the broader clinical context that meaningful diagnosis requires. As a result, the field has progressively shifted from building isolated perceptual tools toward developing integrated systems capable of connecting visual analysis with language and clinical reasoning.

This transition has been accelerated by the emergence of large-scale vision–language models, foundation models trained on diverse multimodal data, and instruction-tuned architectures capable of performing clinical question answering, radiology report generation, and multimodal diagnostic reasoning. These developments represent a qualitative transition in the goals of medical imaging AI—from systems that simply interpret images to systems that can communicate findings and, increasingly, support clinical judgment. Understanding this transition requires a conceptual framework that goes beyond architectural or task-based categorisations to capture the progression of clinical capability that modern systems exhibit.

This paper introduces the Seeing–Saying–Reasoning (SSR) framework, which organises medical imaging AI systems according to the level of clinical intelligence they demonstrate. The three stages of the framework correspond to visual perception of medical images (Seeing), translation of visual information into clinically meaningful language (Saying), and integration of perceptual and linguistic outputs with external medical knowledge to support diagnostic inference (Reasoning). This capability-based perspective provides a structured lens for analysing how current research is advancing medical AI from image interpretation toward clinically actionable intelligence, and for identifying the reliability and safety challenges that arise specifically at each stage of this progression.

The main contributions of this paper are as follows:

- 1) We introduce the SSR framework as a novel capability-based taxonomy for organising and evaluating medical imaging AI systems, accompanied by a formal mathematical definition and a comparison with existing architectural and task-based survey frameworks.

- 2) We present a structured taxonomy of representative multimodal medical AI systems organised according to the SSR framework, covering perception systems, vision–language systems, and emerging clinical reasoning systems.
- 3) We provide a systematic analysis of the reliability and safety challenges that arise at each SSR stage, including hallucination, weak visual grounding, three-dimensional context loss, dataset bias, and evaluation metric limitations.
- 4) We examine the clinical translation and regulatory considerations that govern the path from research prototype to deployed clinical tool, including human–AI collaboration, healthcare infrastructure integration, and post-deployment monitoring.
- 5) We outline a research roadmap linking open challenges to specific SSR stages and identifying the directions most likely to define the next phase of progress in trustworthy medical imaging AI.

The remainder of this paper is organised as follows. Section III surveys related work and positions this paper relative to existing surveys. Section IV traces the historical evolution of AI in medical imaging. Section V reviews the technological foundations underlying multimodal medical AI systems. Section VI introduces the SSR framework formally. Section VII presents the capability-based taxonomy of existing systems. Section VIII analyses reliability and safety challenges. Section IX examines clinical translation and deployment. Section X outlines the research roadmap. Section XI concludes the paper.

II. METHODOLOGY

This survey is based on a narrative review of literature published between 2020 and 2025. Relevant works were identified through searches of Google Scholar, arXiv, and IEEE Xplore using terms including “medical imaging AI,” “multimodal medical AI,” “vision-language models radiology,” “deep learning medical image segmentation,” “radiology report generation,” “visual question answering medical imaging,” and “foundation models healthcare.” Additional papers were identified through backward citation tracing from highly cited works. Studies were selected based on relevance to the three capability stages of the proposed Seeing–Saying–Reasoning framework, with preference given to peer-reviewed publications and influential preprints that have shaped the direction of the field.

III. RELATED WORK

A substantial body of survey literature has examined artificial intelligence in medical imaging, providing important context for the present work. This section positions our contribution relative to the most closely related existing surveys, identifying the gaps that the Seeing–Saying–Reasoning framework is designed to address.

A. Surveys on Deep Learning in Medical Image Analysis

Litjens et al. provided an early and influential survey of deep learning in medical image analysis, reviewing over 300 contributions and surveying the use of deep learning for image classification, object detection, segmentation, registration, and other tasks. Published in 2017, this work established a foundational reference for the deep learning era in medical imaging and remains among the most widely cited papers in the field. However, its scope is necessarily limited to the Seeing stage of the SSR framework—it addresses perceptual models exclusively and predates the emergence of vision–language systems, contrastive pretraining, and multimodal foundation models that now define the frontier of the field.

Shamshad et al. provided the first survey to comprehensively cover applications of transformers in the medical imaging domain, reviewing more than 125 relevant papers across medical image segmentation, detection, classification, reconstruction, synthesis, registration, and clinical report generation. While this survey includes report generation as one application among many, its organising principle remains architectural—transformers versus CNNs—rather than capability-based. It does not address the progression from visual perception to language generation to clinical reasoning as a unified conceptual trajectory, nor does it examine reliability challenges specific to language-enabled systems such as hallucination and weak visual grounding.

B. Surveys on Medical Vision–Language Models

A growing number of surveys have examined vision–language models specifically in the medical domain. Waqas et al. reviewed recent advancements in developing VLMs specialised for healthcare, focusing on publicly available models designed for medical report generation and visual question answering, providing in-depth analyses of architectures and pre-training strategies of sixteen recent noteworthy medical VLMs. Similarly, a systematic review of 115 papers from 2018 to 2025 examined VLM components including fine-tuning strategies, prompt engineering techniques, pre-trained models, adapter modules, and benchmarking datasets, emphasising the role of parameter-efficient adaptation methods in reducing computational overhead while maintaining strong task performance in real-world deployment contexts.

While these surveys provide valuable catalogues of vision–language systems and their technical configurations, they organise their literature primarily according to architecture or task type rather than the level of clinical capability a system achieves. Neither survey provides a principled account of how perception, language, and reasoning represent distinct and progressive stages of clinical intelligence, nor do they examine how the specific reliability challenges introduced at each stage—particularly hallucination, weak grounding, and clinical validation gaps—relate to each other as a coherent problem space.

C. Positioning of the Present Survey

The present survey differs from existing work in three principal respects. First, rather than organising the literature by architecture, modality, or task, we introduce the Seeing–Saying–Reasoning (SSR) framework—a capability-based taxonomy that positions medical imaging AI systems according to the level of clinical intelligence they demonstrate. This perspective reveals a developmental trajectory in the field that architectural or task-based taxonomies obscure. Second, we provide an explicit analysis of the reliability and safety challenges that arise specifically at each SSR stage, treating hallucination, weak grounding, three-dimensional context loss, dataset bias, and evaluation metric limitations as a structured problem space rather than an unrelated collection of concerns. Third, we examine clinical translation and regulatory considerations within the same conceptual framework, connecting technical capabilities to the conditions required for trustworthy clinical deployment. Together, these contributions offer a perspective on the evolution of medical imaging AI that complements and extends the existing survey literature.

IV. EVOLUTION OF AI IN MEDICAL IMAGING

Artificial intelligence in medical imaging has undergone substantial transformation over the past five decades. Beginning with rule-based computer-aided diagnosis systems in the 1970s and 1980s, the field progressed through machine learning, deep learning, and large-scale representation learning, arriving at the current era of multimodal foundation models capable of integrating vision, language, and clinical knowledge. Understanding this progression is essential for contextualising the Seeing–Saying–Reasoning framework introduced in this paper and for appreciating the open challenges that modern systems must address.

A. Early Computer-Aided Diagnosis (1970s–2000s)

Computational approaches to medical imaging date to the 1970s, with dedicated research in computer-aided detection and diagnosis gaining momentum through the 1980s and 1990s. These systems relied on handcrafted features designed to capture clinically relevant image properties—including texture, intensity gradients, shape descriptors, and spatial relationships—combined with classical classifiers such as support vector machines and decision trees. In 1998, the ImageChecker system became the first FDA-approved CAD tool for mammography, marking the transition from laboratory research into regulated clinical use. Similar systems were subsequently developed for lung nodule detection in CT, colon polyp identification, and diabetic retinopathy screening.

Despite demonstrating measurable improvements in radiologist sensitivity under controlled conditions, these systems suffered from high false-positive rates, poor generalisation across scanner vendors and acquisition protocols, and an inability to adapt beyond their design specifications. Their performance was fundamentally bounded by the expressiveness of handcrafted features, which could not capture the full

variability of pathological appearance across diverse patient populations.

B. The Rise of Deep Learning (2012–2018)

A decisive shift followed the success of AlexNet at the ImageNet Large Scale Visual Recognition Challenge in 2012, which demonstrated that learned hierarchical features substantially outperformed handcrafted alternatives. Advances in GPU computing, together with the availability of large annotated datasets, accelerated the adoption of deep learning methods in medical imaging. U-Net, proposed by Ronneberger et al. at MICCAI 2015, represented a particular breakthrough—its encoder-decoder architecture with skip connections enabled precise segmentation from limited annotated data, addressing one of the most persistent constraints in medical AI. U-Net and its descendants became foundational across CT, MRI, histopathology, and retinal imaging, and remain among the most widely cited architectures in the field.

The release of ChestX-ray14, containing over 112,000 labeled chest radiographs, enabled large-scale supervised training for thoracic disease classification. MIMIC-CXR provided paired radiology images and free-text reports, and the BraTS challenge supplied annotated multimodal MRI volumes for brain tumour segmentation—collectively establishing the benchmark infrastructure on which the deep learning era was built. Despite impressive benchmark performance, models of this period remained narrow in scope, optimised for specific tasks and modalities, with limited ability to generalise across institutions or imaging conditions.

C. Representation Learning and Large-Scale Pretraining (2019–2021)

As the limitations of task-specific supervised learning became apparent, the field turned toward methods capable of extracting more general and transferable representations. Inspired by the success of BERT and its domain-adapted variants BioBERT and ClinicalBERT, researchers began exploring pretraining strategies that could reduce dependence on large labeled datasets. Self-supervised contrastive learning emerged as a particularly productive direction. ConVIRT (Zhang et al., 2020) proposed learning medical visual representations from naturally paired images and radiology reports using a bidirectional contrastive objective, requiring only 10% of labeled data to match or exceed the performance of fully supervised ImageNet-initialised models. GLoRIA (Huang et al., 2021) extended this to local image–text alignment, learning correspondences between image subregions and individual words in clinical reports.

These developments marked a conceptual shift: rather than learning from isolated labeled examples, models could now leverage the rich supervision signal contained in clinical text routinely paired with imaging data. This transition laid the groundwork for the vision–language systems that would define the following era.

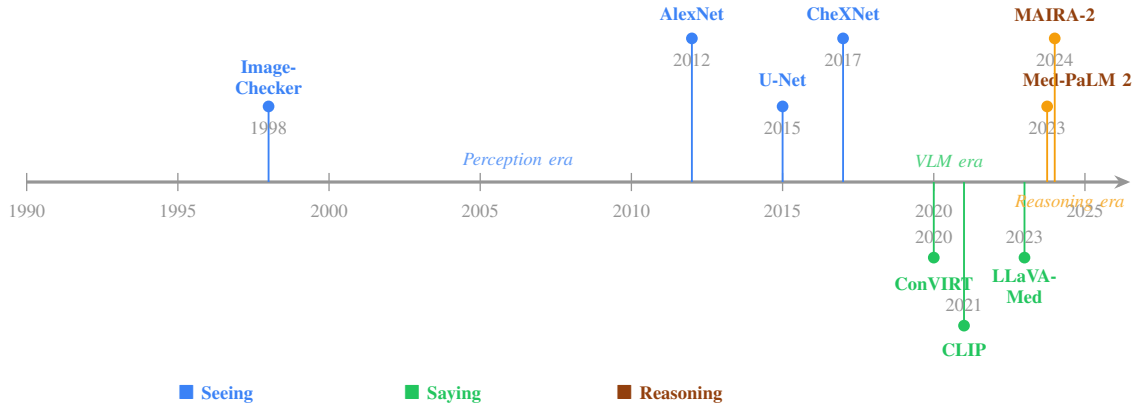


Fig. 1. Timeline of key milestones in medical imaging AI from 1990 to 2025, colour-coded by SSR stage. Blue markers (above axis) denote Seeing-stage advances; green markers (below axis) denote Saying-stage developments; amber markers (above axis) denote Reasoning-stage systems. The progression reflects the field’s transition from perception-focused tools toward language-enabled and reasoning-capable architectures.

D. Emergence of Multimodal and Foundation Models (2021–Present)

The introduction of CLIP by Radford et al. in 2021—trained on 400 million image–text pairs via contrastive learning—catalysed a wave of multimodal research in medical imaging. Domain-adapted successors including MedCLIP, BioViL, and BiomedCLIP demonstrated that CLIP-style pretraining could support zero-shot classification, phrase grounding, and cross-modal retrieval from radiology data. Instruction-tuned multimodal architectures further expanded these capabilities: LLaVA-Med addressed biomedical visual question answering through curriculum learning on PMC-15M, while Med-PaLM 2 achieved expert-level performance on USMLE-style clinical benchmarks. Most recently, generalist models such as Med-PaLM M demonstrated the ability to perform multiple clinical tasks spanning radiology, pathology, genomics, and ophthalmology within a single unified architecture without task-specific fine-tuning.

Regulatory approvals of AI-based diagnostic tools have increased substantially in recent years. As of 2023, the U.S. Food and Drug Administration had cleared over 500 AI-enabled medical devices, the majority targeting radiology applications, reflecting growing confidence in the clinical utility of machine learning systems beyond academic research settings.

E. From Image Analysis to Clinical Intelligence

The developments described above trace a clear trajectory: from handcrafted feature engineering and rule-based detection, through task-specific deep learning and benchmark-driven progress, through generalisable representation learning, toward unified multimodal systems connecting visual perception with language and clinical knowledge. This historical progression naturally aligns with the Seeing–Saying–Reasoning framework introduced in Section VI. Early CAD systems correspond primarily to the Seeing stage, operating as narrowly scoped perceptual tools. Deep learning expanded visual perception capabilities across modalities and tasks, maturing the Seeing

stage substantially. Contrastive and self-supervised representation learning enabled stronger cross-modal alignment underlying the Saying stage, while modern multimodal foundation models increasingly approach the Reasoning stage of clinical AI, integrating perception, language, and knowledge within unified architectures. The open challenges identified at each historical phase—from generalisation failures in early CAD to hallucination in modern vision–language models—remain active areas of research addressed in the sections that follow.

V. FOUNDATIONS OF MULTIMODAL MEDICAL AI

Recent advances in medical imaging AI have been driven by the convergence of large-scale visual learning, multimodal representation learning, and the increasing availability of paired clinical data. Modern systems no longer analyze images in isolation; they increasingly integrate visual information with textual reports, clinical records, and structured medical knowledge. Understanding the technological foundations that enable this integration is important for contextualising the progression from image perception to language generation and clinical reasoning that the SSR framework describes.

A. Medical Imaging Modalities and Their AI Challenges

Medical imaging encompasses a diverse range of modalities, each capturing distinct aspects of human anatomy and physiology. Computed tomography (CT) provides high-resolution cross-sectional views of internal structures and is widely used for chest, abdominal, and neurological diagnosis. Magnetic resonance imaging (MRI) offers superior soft-tissue contrast without ionising radiation, making it the modality of choice for brain, musculoskeletal, and cardiac applications. X-ray imaging remains the most prevalent modality in clinical practice due to its low cost and accessibility, while ultrasound, retinal fundus imaging, and whole-slide histopathology provide complementary diagnostic views across specialties.

From an AI perspective, medical images differ substantially from natural images in ways that complicate model design. Many diagnostic modalities produce volumetric data—CT and

MRI scans consist of hundreds of axial slices capturing three-dimensional anatomy—requiring models capable of processing spatial relationships across multiple planes rather than single frames. Clinically meaningful features are often subtle, context-dependent, and highly variable in appearance across patients, scanner manufacturers, and acquisition protocols. These characteristics motivate the development of specialised architectures, domain-adapted pretraining strategies, and evaluation protocols that go beyond standard natural image benchmarks.

B. Learning Paradigms in Medical Imaging AI

The dominant learning paradigm in early medical imaging AI was fully supervised training on manually annotated datasets. While effective, this approach is severely constrained by the high cost and clinical expertise required to produce high-quality labels at scale. Annotating a single CT volume for organ segmentation may require hours of expert labeling time, limiting the size and diversity of available training corpora.

To address this bottleneck, the field has increasingly adopted self-supervised and contrastive learning strategies that extract useful representations from unlabeled or weakly labeled data. Methods such as masked image modeling—adapted from masked language modeling in NLP—enable models to learn rich visual representations by reconstructing masked image regions without manual annotation. Contrastive approaches such as MoCo and SimCLR, when applied to medical imaging, have demonstrated strong transfer performance across downstream clinical tasks with substantially reduced labeled data requirements. Transformer architectures and vision transformers have further improved representation learning by enabling global context modeling across large imaging volumes. Large-scale multi-task pretraining frameworks further improve generalisation by exposing models to diverse imaging modalities and clinical tasks within a unified training objective, reducing the need for task-specific architectures.

C. Multimodal Fusion Strategies

A defining characteristic of contemporary medical AI is the integration of information from multiple data sources. Multimodal medical AI systems typically combine heterogeneous inputs through one of three principal fusion strategies. Early fusion concatenates raw or lightly processed inputs from different modalities before any deep feature extraction, allowing the model to learn cross-modal interactions from the ground up. Late fusion processes each modality independently through dedicated encoders and combines their representations at the decision level, preserving modality-specific features while enabling complementary prediction. Joint representation learning—the most widely adopted strategy in recent vision-language systems—trains shared embedding spaces in which image and text representations are aligned through contrastive or generative objectives, enabling flexible cross-modal reasoning and retrieval.

In clinical practice, the choice of fusion strategy reflects the nature of the task and the availability of paired training data.

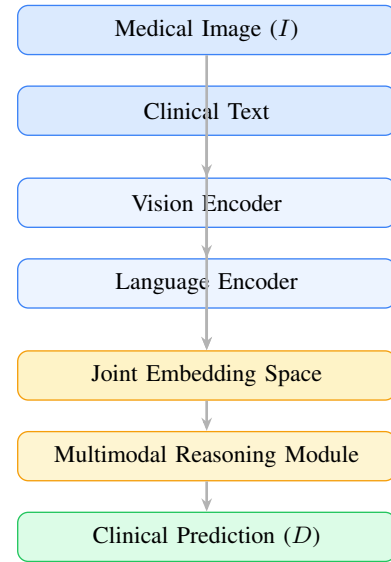


Fig. 2. General architecture of a multimodal medical AI system. A vision encoder and a language encoder independently process imaging and textual inputs and align them in a joint embedding space. A multimodal reasoning module integrates these representations to produce a clinical prediction D .

Systems designed for radiology report generation typically employ joint representation learning between image encoders and language model decoders. Systems supporting clinical decision support may additionally incorporate structured knowledge from electronic health records or medical ontologies through graph-based or attention-based fusion mechanisms. A schematic overview of this architecture is provided in Figure 2.

D. Vision–Language and Multimodal Systems

The emergence of vision–language models represents the most significant architectural development in medical imaging AI in recent years. In clinical practice, images are rarely interpreted in isolation—radiologists rely on written reports, prior examinations, patient histories, and clinical notes to contextualise findings. Vision–language models replicate aspects of this process by learning joint representations of images and text, enabling applications including automated report generation, visual question answering, and cross-modal retrieval. Recent medical vision–language models such as BioViL, MedCLIP, and LLaVA-Med demonstrate how joint image–text representations can support report generation, visual question answering, and cross-modal retrieval across a range of clinical tasks and imaging modalities.

By learning joint representations of images and text, multimodal models enable AI systems to communicate clinical findings and interact with medical knowledge sources. These capabilities represent a transition from purely perceptual models toward systems capable of supporting clinical reasoning, motivating the Seeing–Saying–Reasoning framework introduced in Section VI.

VI. THE SEEING–SAYING–REASONING FRAMEWORK

Recent advances in medical imaging AI have produced a rapidly expanding body of research spanning image analysis, multimodal representation learning, and clinical decision support. Prior surveys have organized this literature primarily according to model architecture, imaging modality, or downstream task. While informative, these categorizations primarily reflect engineering design choices rather than system capability, and often fail to capture the progression of clinical intelligence that modern systems exhibit. As large multimodal foundation models increasingly unify perception, language understanding, and reasoning within a single architecture, traditional taxonomies based on model type or task become less informative. To better contextualize this shift, we introduce the Seeing–Saying–Reasoning (SSR) framework, which organizes medical imaging AI systems according to the level of clinical intelligence and interpretive capability they demonstrate rather than their architectural design. This capability-oriented perspective provides a structured way to analyze how current research advances medical AI from perception toward actionable clinical insight.

A. Formal Definition

Let I denote a medical image, V a learned visual representation, T a clinical textual description, K external medical knowledge, and D the resulting diagnostic inference or clinical decision-support output. The SSR framework conceptualizes medical imaging AI as a progression of functional mappings:

$$\textbf{Seeing: } V = f(I) \quad (1)$$

$$\textbf{Saying: } T = g(V) \quad (2)$$

$$\textbf{Reasoning: } D = h(I, T, K) \quad (3)$$

This formulation emphasizes that each stage transforms information into progressively richer clinical representations, enabling the transition from perceptual analysis to knowledge-informed decision support. Importantly, the three stages are hierarchical rather than independent—systems operating at the Reasoning stage rely on accurate visual perception and well-grounded language generation, making progress in earlier stages a prerequisite for reliable clinical reasoning.

Conceptually, the SSR framework mirrors the cognitive workflow of clinical interpretation: clinicians first perceive imaging patterns, then describe findings in language, and finally integrate those findings with medical knowledge to form diagnostic conclusions. This alignment with human clinical reasoning provides both an intuitive basis for the framework and a principled standard against which AI system capabilities can be evaluated.

The three-stage progression is illustrated in Figure 3.

B. Comparison With Existing Surveys

Prior surveys of medical imaging AI typically categorize models according to architecture—CNNs versus transformers—or application domain, such as segmentation,

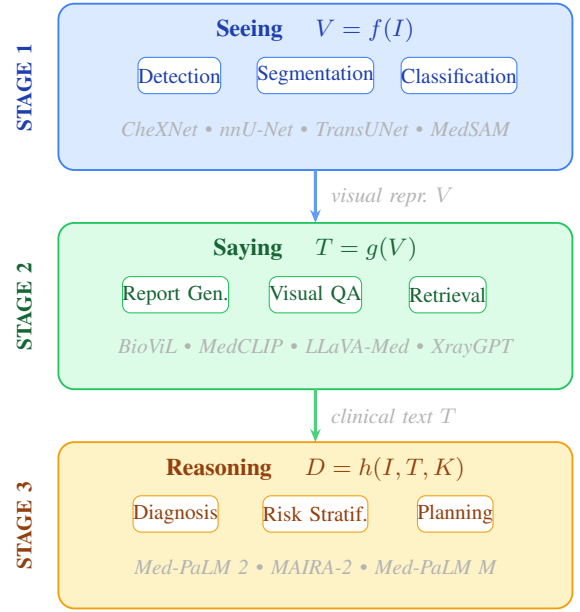


Fig. 3. The Seeing–Saying–Reasoning (SSR) framework organises medical imaging AI systems as a hierarchical progression of three functional stages. *Seeing* extracts a visual representation V from a medical image I . *Saying* translates V into a clinical textual description T . *Reasoning* integrates I , T , and external knowledge K to produce a diagnostic inference D . Representative systems are shown for each stage.

detection, or classification. While these perspectives are useful for tracking technical progress, they do not capture the functional evolution of what AI systems can contribute clinically. The SSR framework addresses this gap by emphasizing the progression from visual perception to multimodal communication and ultimately clinical reasoning. Rather than asking what architecture a system uses, SSR asks what clinical role a system can fulfill—a distinction that becomes increasingly important as foundation models blur traditional architectural boundaries.

C. Seeing: Visual Perception of Medical Images

The first stage, *Seeing*, encompasses AI systems designed to interpret medical images and extract clinically relevant visual features. Tasks include lesion detection, organ segmentation, abnormality classification, and pathological structure localization across modalities such as CT, MRI, X-ray, histopathology, and retinal imaging. Early progress was driven primarily by convolutional neural networks (CNNs) trained on labeled datasets, while more recent work leverages transformer-based architectures and large-scale pretrained vision models.

Representative systems include CheXNet, a DenseNet-121-based model demonstrating radiologist-level performance on chest X-ray pathology detection across fourteen disease categories, and nnU-Net, a self-configuring segmentation framework that ranked first in nineteen international challenges across diverse imaging modalities. Transformer-based approaches such as TransUNet and Swin-UNet subsequently improved segmentation by capturing long-range spatial dependencies. More recently, prompt-based foundation models such

as the medical adaptation of the Segment Anything Model (SAM) have demonstrated the ability to generalize across modalities with minimal fine-tuning.

Despite these advances, Seeing-stage systems remain limited in scope. They produce structured outputs—bounding boxes, segmentation masks, or classification scores—that require further clinical interpretation and do not communicate findings in natural language, highlighting the need for systems that can communicate and contextualize visual findings within broader clinical workflows.

D. Saying: Connecting Vision and Medical Language

The second stage, *Saying*, refers to systems that translate visual information into textual descriptions or structured clinical narratives. Applications include automated radiology report generation, visual question answering, cross-modal retrieval, and contrastive vision–language pretraining.

BioViL combines a CNN image encoder with a BERT text encoder through contrastive learning on MIMIC-CXR, achieving strong performance on disease classification and phrase grounding. MedCLIP replaces the standard contrastive loss with a semantic matching objective, improving data efficiency across unpaired image–text pairs. LLaVA-Med adapts the LLaVA architecture to the biomedical domain through curriculum learning on PMC-15M—a dataset of fifteen million biomedical figure-caption pairs—achieving competitive results on VQA-RAD, SLAKE, and PathVQA benchmarks.

The integration of language introduces reliability challenges that are largely absent from purely perceptual systems, including hallucinated findings, incomplete descriptions, and weak grounding between visual evidence and generated text—challenges addressed in detail in Section VIII. While *Saying*-stage systems enable AI to communicate observations in natural language, they do not perform clinical inference. The transition to Reasoning therefore involves integrating perceptual outputs and textual representations with broader medical knowledge to generate structured clinical conclusions.

E. Reasoning: Toward Clinical Interpretation

The third stage, *Reasoning*, represents AI frameworks that move beyond perception and description toward supporting higher-level clinical interpretation. These systems integrate imaging information with electronic health records, diagnostic guidelines, prior examinations, and medical literature to assist with diagnostic suggestion, risk stratification, and treatment planning.

Med-PaLM 2 achieved expert-level performance on USMLE-style clinical questions and demonstrated multimodal reasoning over medical images and clinical text. MAIRA-2 advances the *Saying*-to-Reasoning transition by generating grounded radiology reports that associate textual findings with specific image regions, enabling direct clinician verification of AI-generated conclusions. Evaluations of GPT-4V in clinical settings further indicate that large multimodal architectures can perform diagnostic reasoning across radiology, pathology, and ophthalmology in zero-shot settings, though with inconsistent

accuracy that underscores the gap between general capability and reliable deployment.

Unlike perception tasks, clinical reasoning requires integrating heterogeneous evidence while maintaining transparency and safety. Current multimodal models demonstrate promising capabilities but often lack the reliability and interpretability required for real-world clinical deployment. Reliable clinical reasoning therefore remains an open problem and arguably the central frontier of medical imaging AI research. Bridging the gap between descriptive multimodal models and clinically validated reasoning systems will likely define the next phase of progress in medical AI.

Building on this conceptual framework, Section VII introduces a taxonomy of multimodal medical AI systems categorized according to the stage of capability they primarily address.

VII. TAXONOMY OF MULTIMODAL AI SYSTEMS IN MEDICAL IMAGING

Building on the SSR framework introduced in Section VI, this section presents a structured taxonomy of representative multimodal AI systems in medical imaging. Rather than cataloguing models by architecture alone, we organize existing work according to the stage of clinical capability each system primarily addresses. Systems are selected on the basis of their influence, methodological significance, and breadth of evaluation across publicly available benchmarks. For each entry, we describe the model’s core contribution, underlying architecture, training data, and primary limitation. A summary comparison of all systems is provided in Table I.

A. Perception Systems (Seeing)

Perception systems form the foundation of medical imaging AI pipelines. These models analyze visual data to detect, classify, or segment anatomical structures and pathological findings, and constitute the primary instantiation of the Seeing stage in the SSR framework.

CheXNet (Rajpurkar et al., 2017) is a 121-layer DenseNet trained on the NIH ChestX-ray14 dataset, which contains over 112,000 frontal-view chest X-ray images labeled across fourteen pathology classes. The model leverages dense connectivity to improve gradient flow and feature reuse, enabling effective learning from large-scale radiographic data. CheXNet achieved an F_1 score of 0.435 on pneumonia detection, surpassing the average radiologist score of 0.387, and set state-of-the-art performance across all fourteen pathologies at the time of publication. A key limitation is its restriction to a single imaging modality and the absence of any language-based output, confining its utility to classification and localization tasks.

nnU-Net (Isensee et al., 2021) is a self-configuring segmentation framework published in *Nature Methods* that automatically adapts its architecture, preprocessing pipeline, and training strategy to any new dataset without manual intervention. By encoding expert design knowledge as a set of fixed parameters and empirical heuristics, nnU-Net surpassed specialized

solutions on 23 public datasets used in international biomedical segmentation competitions, and subsequent versions have continued to underpin winning entries at MICCAI challenges. Its primary limitation is that the automation process offers limited interpretability, and manually tuned approaches may outperform it on highly specialized tasks.

TransUNet (Chen et al., 2021) integrates Vision Transformer (ViT) encoders into a U-Net segmentation backbone, combining the local feature extraction capabilities of CNNs with the global context modelling of self-attention. Evaluated on the Synapse multi-organ CT segmentation benchmark, TransUNet achieved a mean Dice score of 77.48%, outperforming CNN-only baselines. The model demonstrated that transformer-based architectures can capture long-range spatial dependencies critical to accurate organ delineation. Its limitation is high computational cost relative to CNN baselines and reduced performance on datasets too small to support large-scale pretraining.

Medical SAM (Ma et al., 2024) adapts the Segment Anything Model (SAM) to the medical domain through lightweight fine-tuning on a large collection of annotated medical image volumes. By supporting prompt-based segmentation via point, box, and mask inputs, MedSAM enables flexible generalization across CT, MRI, ultrasound, and histopathology with minimal task-specific adaptation. A key limitation is that performance on complex three-dimensional structures and rare pathologies remains sensitive to prompt quality and may require additional domain-specific fine-tuning.

B. Vision–Language Systems (Saying)

Vision–language systems extend perception models by connecting visual representations with natural language, enabling applications such as automated report generation, visual question answering, and cross-modal retrieval. These systems constitute the primary instantiation of the Saying stage.

BioViL (Bannur et al., 2023) combines a CNN-based image encoder with a BERT-based text encoder trained through local contrastive learning on paired radiology images and reports from the MIMIC-CXR dataset. By learning fine-grained alignment between image regions and clinical phrases, BioViL achieves strong performance on phrase grounding and disease classification tasks while producing representations that support cross-modal retrieval. Its primary limitation is restriction to the chest radiology domain, which constrains generalization across other imaging modalities.

MedCLIP (Wang et al., 2022) adapts the CLIP contrastive learning framework to the medical domain by replacing the standard InfoNCE loss with a semantic matching objective grounded in medical knowledge graphs. This approach decouples image–text pairing from strict co-occurrence, improving data efficiency and enabling learning from unpaired medical images and reports. MedCLIP demonstrated competitive performance on zero-shot classification and retrieval tasks across chest X-ray benchmarks. A limitation is that its grounding capability remains weaker than models trained with explicit region-level supervision.

LLaVA-Med (Li et al., 2023) adapts the general-purpose LLaVA vision–language model to the biomedical domain through a two-stage curriculum learning strategy. In the first stage, the model is pretrained on PMC-15M, a dataset of fifteen million figure-caption pairs from biomedical literature. In the second stage, it is fine-tuned using a self-generated instruction-following dataset. LLaVA-Med achieved competitive performance on the VQA-RAD, SLAKE, and PathVQA benchmarks and was accepted at NeurIPS 2023. A central limitation is that the model is not intended for direct clinical deployment, and hallucinated responses remain a documented concern.

XrayGPT (Thawkar et al., 2023) combines the MedCLIP visual encoder with a fine-tuned Vicuna large language model to enable interactive radiology report generation and clinical question answering from chest X-ray images. The model supports open-ended dialogue about radiological findings, representing an early step toward conversational clinical AI. Its limitation is restriction to the chest radiology domain and a reliance on the quality of MedCLIP visual representations, which may introduce propagated errors in report generation.

C. Clinical Reasoning Systems (Reasoning)

Clinical reasoning systems represent the most advanced category in the SSR taxonomy. These models integrate imaging data with external medical knowledge, clinical text, or structured records to support diagnostic inference and decision-making, instantiating the Reasoning stage of the framework.

Med-PaLM 2 (Singhal et al., 2023) is a large language model developed by Google, built on the PaLM 2 foundation and fine-tuned using medical domain knowledge and an ensemble refinement prompting strategy. Med-PaLM 2 was the first model to reach expert-level performance on USMLE-style questions, achieving an accuracy of 86.5% on the MedQA benchmark. In human evaluation across 1,066 consumer medical questions, physicians preferred Med-PaLM 2 answers over those of other physicians on eight of nine clinical utility axes. A key limitation is that the model is closed-source and has not been validated in prospective real-world clinical settings.

MAIRA-2 (Bannur et al., 2024) is a multimodal model developed by Microsoft Research for grounded radiology report generation. Unlike prior report generation systems that produce free-text outputs disconnected from specific image regions, MAIRA-2 associates each generated finding with a corresponding bounding box in the source image, enabling direct clinician verification of AI-generated content. The model was evaluated on the MIMIC-CXR dataset and demonstrated improvements in both report quality and grounding precision over prior baselines. Its limitation is restriction to chest radiology, and bounding box quality varies across pathology types and image conditions.

Med-PaLM M (Tu et al., 2023) extends the Med-PaLM framework to multiple medical modalities including chest X-ray, mammography, retinal imaging, histopathology, and genomics within a single unified architecture. Built on the

PaLM-E vision–language model, Med-PaLM M demonstrated generalist performance across medical question answering, visual question answering, and report summarization without task-specific fine-tuning. This generalist capability represents a significant step toward the Reasoning stage of the SSR framework, though the model’s performance on specialized reasoning tasks remains below that of domain-specific systems.

D. Taxonomy Summary

The three categories described above illustrate how medical imaging AI systems are evolving across increasing levels of clinical capability. Perception models provide accurate visual analysis but are limited to structured outputs requiring clinical interpretation. Vision–language systems enable communication of findings in natural language but remain prone to reliability issues including hallucination and weak grounding. Emerging reasoning systems integrate heterogeneous data sources to support clinical decision-making, but face substantial challenges related to validation, interpretability, and safety.

Importantly, these categories are not mutually exclusive. Several recent systems combine elements from multiple stages: MAIRA-2 bridges Saying and Reasoning by grounding language in visual evidence, while Med-PaLM M spans all three stages within a unified generalist architecture. This trend toward integrated multimodal intelligence reflects the broader direction of the field and underscores the value of a capability-oriented taxonomy. A comparative summary of all systems discussed in this section is provided in Table I.

VIII. RELIABILITY AND SAFETY CHALLENGES

Despite rapid progress in medical imaging AI, reliability remains a central concern that constrains real-world clinical adoption. The transition from Seeing-stage perception models to Saying and Reasoning-stage systems introduces qualitatively new failure modes—a model that misclassifies a lesion produces a discrete error, whereas a model that generates a fluent but clinically incorrect radiology report may mislead a physician without triggering obvious suspicion. Each stage of the SSR framework therefore carries distinct reliability risks: Seeing-stage systems are primarily affected by dataset bias and loss of three-dimensional context; Saying-stage systems introduce hallucination and weak visual grounding; and Reasoning-stage systems face the additional challenges of heterogeneous data integration, interpretability, and validated clinical deployment. This section examines each of these challenges in detail and surveys current research directions aimed at addressing them.

A. Hallucination in Vision–Language Models

Hallucination—the generation of text unsupported by visual evidence or clinical facts—is among the most consequential reliability failures in Saying-stage medical AI systems. In healthcare, a visual assistant that hallucinates can cause misdiagnoses or inappropriate treatments. Two principal forms of hallucination have been identified in the literature. *Object*

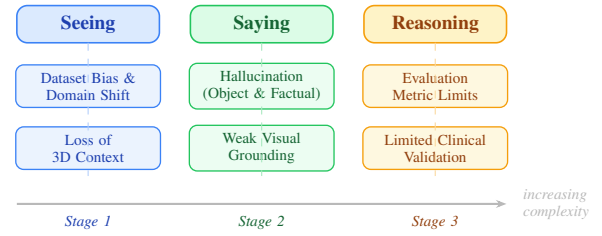


Fig. 4. Reliability and safety challenges organised by SSR stage. Each column lists the primary failure modes arising as systems progress from visual perception (*Seeing*) through language generation (*Saying*) to clinical reasoning (*Reasoning*).

hallucination refers to the description of findings that are not present in the image—for example, reporting a pulmonary nodule in a clear chest radiograph. *Factual hallucination* refers to the generation of plausible but clinically incorrect statements, such as attributing a finding to the wrong anatomical location or mischaracterising its severity.

Empirical evaluations have confirmed that hallucination is a significant and persistent problem across current medical vision–language models. The MedVH benchmark, introduced to evaluate hallucination in domain-specific large vision–language models, found that medical LVLs are particularly susceptible to hallucinations, often more so than general-domain models, despite demonstrating promising performance on standard medical tasks. LLaVA-Med, despite strong performance on biomedical question answering benchmarks, was explicitly acknowledged to risk producing incorrect responses or complete hallucinations on biomedical questions. Proposed mitigation strategies include retrieval-augmented generation, which grounds model outputs in retrieved clinical evidence, region-aware training objectives that enforce stricter alignment between textual claims and image regions, and uncertainty-aware decoding strategies that allow models to express confidence rather than generating fluent but unsupported statements.

B. Weak Grounding Between Images and Text

Closely related to hallucination is the problem of weak grounding—the failure to explicitly link specific textual statements to identifiable visual evidence in the source image. In many vision–language systems, the relationship between visual features and generated text is implicit: models learn statistical associations between diagnostic phrases and image patterns rather than constructing verifiable correspondences between textual claims and image regions. While such associations can yield strong benchmark performance, they undermine interpretability and clinical trust.

In clinical environments, a radiologist reviewing an AI-generated report needs to verify that each reported finding corresponds to a specific and locatable region of the image. Systems that cannot provide this correspondence produce outputs that are difficult to validate and therefore difficult to trust. MAIRA-2 represents one of the most direct attempts to address this limitation by generating grounded radiology reports that associate each textual finding with a bounding box region

TABLE I
TAXONOMY OF REPRESENTATIVE MULTIMODAL AI SYSTEMS IN MEDICAL IMAGING, ORGANIZED BY SSR STAGE. YEAR REFERS TO INITIAL PUBLICATION OR PREPRINT RELEASE.

Model	Stage	Architecture	Modality	Dataset	Key Task	Year
CheXNet	Seeing	DenseNet-121	Chest X-ray	ChestX-ray14	Classification	2017
nnU-Net	Seeing	Self-config. U-Net	Multi-modal	23 challenge sets	Segmentation	2021
TransUNet	Seeing	CNN + ViT	CT	Synapse	Segmentation	2021
Medical SAM	Seeing	ViT (SAM)	Multi-modal	SA-Med2D	Prompt segmentation	2024
BioViL	Saying	CNN + BERT	Chest X-ray	MIMIC-CXR	Report grounding	2023
MedCLIP	Saying	CLIP (semantic loss)	Chest X-ray	CheXpert / MIMIC	Zero-shot retrieval	2022
LLaVA-Med	Saying	LLaVA + PMC-15M	Multi-modal	VQA-RAD / SLAKE	Visual QA	2023
XrayGPT	Saying	MedCLIP + Vicuna	Chest X-ray	OpenI / MIMIC	Report generation	2023
Med-PaLM 2	Reasoning	PaLM 2 (fine-tuned)	Text + Image	MedQA / MultiMedQA	Clinical QA	2023
MAIRA-2	Reasoning	Multimodal LLM	Chest X-ray	MIMIC-CXR	Grounded report gen.	2024
Med-PaLM M	Reasoning	PaLM-E (generalist)	Multi-modal	MultiMedBench	Generalist medical AI	2023

in the source image, enabling direct clinician verification. Approaches to improving grounding more broadly include region-aware contrastive pretraining, explicit visual grounding modules that link noun phrases to image coordinates, and attention-based interpretability tools such as GradCAM and attention rollout that provide post-hoc visualisation of which image regions influenced specific textual outputs.

C. Loss of Three-Dimensional Context

A fundamental challenge in medical imaging AI stems from the volumetric nature of many clinical modalities. CT and MRI examinations produce three-dimensional image volumes consisting of hundreds of axial slices, within which pathological findings must be understood in terms of their spatial extent, location relative to anatomical landmarks, and relationships across planes. A significant proportion of existing deep learning models, however, process these volumes as collections of independent two-dimensional slices, effectively discarding the spatial continuity that is central to accurate clinical interpretation.

This simplification introduces concrete clinical risks. Slice-by-slice processing can produce inconsistent predictions across adjacent slices, misestimate lesion volume and extent, and miss findings—such as pulmonary emboli or subtle white matter lesions—that are only apparent in multi-planar context. For clinical tasks with explicit three-dimensional requirements, including tumour staging, surgical planning, and longitudinal change detection, these limitations are particularly consequential. Architectural responses include 3D U-Net and V-Net variants that extend convolutional segmentation to volumetric inputs, transformer-based approaches such as Med3DVLM that apply self-attention across spatial dimensions, and patch-based strategies that process overlapping three-dimensional windows to balance contextual coverage against computational cost.

D. Dataset Bias and Domain Shift

Medical datasets are inevitably shaped by the institutional contexts in which they are collected—including imaging hardware, acquisition protocols, patient demographics, and clinical

practice norms. Models trained on such datasets may achieve strong in-distribution performance while failing substantially when deployed in new environments, a phenomenon commonly referred to as domain shift. The well-documented performance degradation observed when models trained on MIMIC-CXR are evaluated on CheXpert—two large chest radiograph datasets collected at different institutions—illustrates how even superficially similar datasets can produce significant distributional discrepancies.

Bias in medical AI manifests along multiple dimensions. Models trained on datasets with demographic imbalances may exhibit differential performance across sex, age, or ethnicity. Systems optimised for common pathologies may perform poorly on rare conditions underrepresented in training data. Equipment-induced variation—differences in field strength, scanner manufacturer, or reconstruction kernel—can produce systematic image appearance shifts that degrade model performance independently of the clinical content. Federated learning has emerged as a principal technical strategy for addressing domain shift, enabling collaborative model training across multiple institutions without centralising sensitive patient data. Data augmentation strategies designed to simulate acquisition variability, domain adaptation techniques including adversarial training, and multi-centre validation protocols represent complementary approaches to improving cross-institutional robustness.

E. Limitations of Current Evaluation Metrics

The evaluation of medical AI systems—particularly those operating at the Saying and Reasoning stages—presents fundamental measurement challenges. Standard natural language generation metrics including BLEU and ROUGE assess surface-level lexical overlap between generated and reference texts, a property that bears limited relationship to clinical accuracy. Two radiology reports may receive similar BLEU scores while conveying substantially different diagnostic conclusions; conversely, a clinically accurate report written in different phrasing from the reference may receive a low score.

The inadequacy of standard metrics has motivated the development of clinically grounded alternatives. RadGraph F1

and RadCliQ were proposed as radiology report evaluation metrics demonstrating stronger correlation with radiologist assessments of clinical errors than standard lexical metrics such as BLEU and BERTScore. More recently, GREEN—a generative radiology report evaluation metric—demonstrated a correlation of 0.63 with expert radiologist error counts, outperforming RadGraph F1 despite the latter being trained on human annotations, and approaching the inter-expert correlation range of 0.48 to 0.64. Despite these advances, no single automated metric has achieved sufficient alignment with clinical judgment to serve as a reliable proxy for expert evaluation. Clinician involvement in evaluation pipelines, region-level localisation metrics, and task-specific diagnostic accuracy assessments remain necessary complements to automated scoring.

F. Limited Real-World Clinical Validation

A pervasive gap between research performance and clinical reliability stems from the conditions under which medical AI systems are developed and tested. Benchmark datasets, while essential for advancing the field, are curated under controlled conditions that do not fully represent the heterogeneity of real clinical environments. Differences in patient acuity, imaging indication, workflow integration, and operator variability introduce sources of performance variation that controlled benchmarks cannot capture.

Prospective clinical validation—in which models are evaluated under real deployment conditions, with real patients, in real workflows—remains rare in the medical AI literature. The IDx-DR system for autonomous diabetic retinopathy screening, which received FDA De Novo authorisation in 2018 following a prospective study across ten primary care sites, represents one of the few examples of a medical AI system supported by prospective multi-centre clinical evidence at the time of regulatory submission. More broadly, without large-scale prospective studies demonstrating preserved performance across diverse patient populations and healthcare settings, the clinical impact of even technically strong systems remains uncertain. Bridging the gap between experimental success and reliable clinical deployment therefore represents not only a technical challenge but also an institutional and regulatory one that requires sustained collaboration between researchers, clinicians, and healthcare systems.

G. Summary

Table II summarises the reliability challenges discussed in this section, organised according to the SSR stage each challenge primarily affects and the research directions being pursued to address them. While Seeing-stage systems face relatively well-understood challenges related to data diversity and three-dimensional representation, the transition to language-enabled and reasoning-capable systems introduces a qualitatively richer set of failure modes. Addressing these challenges in a principled and clinically grounded manner is a prerequisite for the trustworthy deployment of medical imaging AI.

TABLE II
SUMMARY OF RELIABILITY CHALLENGES IN MULTIMODAL MEDICAL AI, ORGANISED BY SSR STAGE AND PROPOSED MITIGATION DIRECTIONS.

Challenge	SSR Stage	Mitigation directions
Dataset bias & domain shift	Seeing	Federated learning, domain adaptation, diverse datasets
Loss of 3D context	Seeing	3D U-Net, V-Net, Med3DVLM, patch-based 3D transformers
Hallucination	Saying	RAG, region-aware training, uncertainty-aware decoding
Weak visual grounding	Saying	Grounded pretraining, GradCAM, MAIRA-2 bounding boxes
Evaluation metric limitations	Saying / Reasoning	RadGraph F1, GREEN, RadCliQ, clinician-in-the-loop evaluation
Limited clinical validation	Reasoning	Prospective studies, multi-centre trials, regulatory frameworks

IX. CLINICAL TRANSLATION AND DEPLOYMENT

The successful integration of medical imaging AI into clinical practice requires addressing challenges that extend well beyond algorithmic performance. Even systems that demonstrate strong results on benchmark datasets may fail to deliver comparable value in real clinical environments, where patient populations are heterogeneous, imaging conditions vary, and clinical workflows impose practical constraints that curated research settings do not capture. Translating research prototypes into trustworthy clinical tools therefore demands careful attention to the human factors of deployment, the infrastructure requirements of healthcare institutions, and the evolving regulatory frameworks that govern medical AI. This section examines these dimensions in an integrated manner, addressing how AI systems must be designed to work alongside clinicians, how they must interface with clinical infrastructure, and what regulatory obligations govern their path to deployment.

A. Human–AI Collaboration and the Decision Support Paradigm

A foundational principle of clinical medical AI is that systems should augment rather than replace human expertise. In radiology and pathology workflows, AI models are typically deployed as decision support tools—flagging suspicious findings, prioritising urgent cases, or generating draft reports that clinicians review and refine. This collaborative model reflects an important empirical reality: while AI systems excel at processing large volumes of imaging data consistently and rapidly, clinicians provide contextual judgment, patient-specific knowledge, and accountability that current AI systems cannot replicate.

The clinical value of this collaborative paradigm has been demonstrated in prospective studies. In the clinical study supporting FDA authorisation of Paige Prostate, 16 pathologists examined 527 prostate biopsy slide images with and without AI assistance. The software improved cancer detection by

7.3% on average, with a 70% reduction in false negative diagnoses and a 24% reduction in false positive diagnoses. Notably, non-specialist pathologists using Paige Prostate achieved diagnostic accuracy equivalent to prostate specialists who were not using the software—a finding that illustrates how AI-assisted workflows can reduce the performance gap between specialists and generalists, with direct implications for healthcare equity and access.

Human-in-the-loop architectures formalise this collaboration by ensuring that AI outputs are reviewed by clinicians before influencing patient care decisions. This oversight is particularly important for complex or ambiguous cases, where models are most likely to encounter out-of-distribution inputs that push performance below benchmark levels. Maintaining clinician involvement not only mitigates patient safety risks but also enables the collection of correction signals that can improve model performance over time through active learning or feedback-driven fine-tuning.

B. Integration with Clinical Infrastructure

For AI systems to deliver clinical value, they must integrate seamlessly with the information systems through which clinical care is delivered. Medical images in hospital environments are typically stored, retrieved, and displayed through Picture Archiving and Communication Systems (PACS) using the DICOM standard, while patient records are managed through Electronic Health Record (EHR) platforms. AI tools must be compatible with these systems to fit naturally into clinical workflows without requiring clinicians to adopt new interfaces or interrupt established routines.

Interoperability standards play a central role in enabling this integration. HL7 FHIR (Fast Healthcare Interoperability Resources) has emerged as a widely adopted standard for exchanging clinical data between AI systems and EHR platforms, enabling AI-generated findings to be surfaced within existing clinical documentation workflows. Proposals for AI-specific extensions to the HL7 FHIR standard have been developed to embed AI compliance metadata—including model identity, inference confidence, and interpretability information—directly within clinical documents, ensuring that AI contributions to clinical decisions are transparent and auditable.

Beyond technical compatibility, integration requires attention to usability, data privacy, and system reliability. If an AI tool introduces additional steps, generates excessive alerts, or disrupts the pacing of a radiologist's reading workflow, adoption will remain limited regardless of technical performance. Workflow studies evaluating the impact of AI integration on reading time, alert fatigue, and diagnostic confidence are therefore an important and underserved area of clinical AI research.

C. Regulatory Frameworks and Approval Pathways

Medical AI systems that support diagnosis or treatment decisions are classified as medical devices in most major jurisdictions and must undergo regulatory review before clinical deployment. In the United States, the FDA evaluates AI-

based medical devices through the De Novo pathway for novel device types and the 510(k) premarket notification pathway for devices substantially equivalent to previously cleared products. Paige Prostate Detect received Class II designation through the FDA's De Novo classification pathway in 2021, establishing a new regulatory category for AI-based pathology tools and creating a 510(k) predicate for subsequent similar products.

In the European Union, the EU Artificial Intelligence Act, which entered into force in August 2024, classifies AI-based software intended for medical purposes as high-risk, imposing requirements for risk-mitigation systems, high-quality datasets, clear user information, and human oversight. Medical device AI qualifies as high-risk under the Act if it is a safety component of a device or constitutes a medical device in itself and is subject to third-party conformity assessment by a notified body. These regulatory developments reflect a broader international trend toward more structured governance of medical AI, with implications for how systems are designed, validated, and monitored throughout their deployment lifecycle.

A distinctive regulatory challenge arises from the adaptive nature of machine learning systems. Unlike traditional medical devices with fixed functionality, AI models may be updated as new data becomes available, potentially altering their behaviour in ways that affect safety. Regulatory agencies are actively developing frameworks for managing this challenge, including approaches for defining the boundaries of acceptable performance change that do not require full re-review.

D. Post-Deployment Monitoring and Continuous Evaluation

Regulatory approval and clinical deployment do not represent the end of a medical AI system's validation lifecycle. Once deployed, systems must be continuously monitored to ensure that their performance remains reliable as clinical populations, imaging technology, and institutional practices evolve. Distribution shifts—arising from changes in patient demographics, scanner upgrades, or modifications to clinical protocols—can silently degrade model accuracy without triggering obvious failures.

Effective post-deployment monitoring requires establishing mechanisms for tracking system performance against ground truth labels, detecting anomalous prediction patterns, and triggering revalidation when performance degradation is identified. Integrating these mechanisms into hospital information systems requires collaboration between AI developers, clinical informaticians, and hospital IT departments. Continuous monitoring also contributes to the accumulation of real-world evidence that supports regulatory submissions, reimbursement applications, and clinician trust—making it not only a safety requirement but a strategic asset for the responsible deployment of medical AI.

E. Summary

Translating medical imaging AI from research prototypes into clinical tools requires simultaneous progress across human factors, technical infrastructure, regulatory compliance, and ongoing monitoring. The evidence reviewed in this section

indicates that when AI systems are designed as decision support tools, validated through prospective clinical studies, integrated with existing clinical infrastructure, and monitored continuously after deployment, they can deliver measurable improvements in diagnostic accuracy and workflow efficiency. Achieving this at scale, across diverse healthcare settings and regulatory jurisdictions, remains one of the defining challenges of the field.

X. RESEARCH ROADMAP FOR TRUSTWORTHY MEDICAL AI

Despite rapid progress in medical imaging AI, several fundamental challenges must be addressed before multimodal systems can be reliably integrated into routine clinical practice. The SSR framework introduced in this paper provides a useful lens for organising these challenges: limitations at the Seeing stage constrain the quality of perceptual inputs on which all higher-stage capabilities depend; limitations at the Saying stage undermine the reliability of clinical communication; and limitations at the Reasoning stage impede the transition to validated decision support. Addressing these issues requires coordinated advances across model design, data infrastructure, evaluation methodology, and human–AI collaboration. This section outlines five research directions that are likely to define the next phase of progress in trustworthy multimodal medical AI.

A. Robust Multimodal Learning Across Heterogeneous Data Sources

Primarily addresses: Reasoning stage

Clinical diagnostic decisions are inherently multimodal. Physicians routinely integrate imaging data with laboratory results, patient history, prior examinations, genomic profiles, and clinical notes when forming diagnostic conclusions. Current AI systems, however, largely process imaging data in isolation, with limited ability to incorporate the heterogeneous non-imaging information that contextualises visual findings. Developing architectures capable of robust fusion across these diverse data types remains one of the most consequential open problems in medical AI.

A scoping review analysing 432 multimodal AI papers published between 2018 and 2024 found that multimodal AI models consistently outperform their unimodal counterparts, with an average improvement of 6.2 percentage points in AUC—a finding that quantifies the clinical value of multimodal integration while underscoring the gap between current systems and the full multimodal integration that clinical practice requires. Promising architectural directions include transformer-based models and graph neural networks capable of jointly processing clinical notes, imaging data, and genomic information, as well as knowledge-graph-augmented fusion strategies that incorporate structured medical ontologies to constrain and contextualise model reasoning. Ensuring that such systems remain interpretable and resistant to spurious cross-modal correlations represents a critical accompanying research challenge.

B. Improving Reliability and Reducing Hallucination

Primarily addresses: Saying stage

As discussed in Section VIII, hallucination represents the most immediate patient safety risk associated with Saying-stage medical AI systems. A systematic evaluation of eleven foundation models across seven medical hallucination tasks found that general-purpose models achieved significantly higher proportions of hallucination-free responses than medical-specialised models, with a median difference of 25.2 percentage points. This counterintuitive finding suggests that domain-specific fine-tuning, as currently practised, may inadvertently increase hallucination risk—a problem that demands new training paradigms rather than incremental refinements to existing approaches.

Promising research directions include retrieval-augmented generation architectures that ground model outputs in verified clinical evidence sources, constitutional AI approaches that enforce factual consistency through model self-critique, and uncertainty-aware output mechanisms that allow systems to express calibrated confidence rather than generating fluent but unsupported statements. Equally important is the development of evaluation frameworks capable of detecting subtle factual hallucinations—cases where generated text uses clinically valid terminology but describes incorrect anatomical locations, wrong laterality, or mischaracterised severity—which current automated metrics are poorly equipped to identify.

C. Learning from Diverse and Distributed Medical Data

Primarily addresses: Seeing and Saying stages

The generalisation failures documented across medical AI systems stem in large part from the limited diversity of available training data. Most benchmark datasets are collected from a small number of academic medical centres, predominantly in high-income countries, using a narrow range of imaging equipment and protocols. Critical challenges in medical multimodal large language model development include handling domain-specific and specialised medical data, limited generalisation in low-data scenarios, and multimodal imbalance—with radiology and pathology image–text pairs substantially overrepresented compared to other critical modalities such as genomic–imaging fusion and multi-omics integration.

Federated learning has emerged as a technically mature strategy for expanding training data diversity without centralising sensitive patient records. The integration of federated learning with vision–language models enables both visual and textual data to be processed in a privacy-preserving, distributed learning environment, allowing models to learn from diverse geographic and demographic distributions without transferring sensitive data. Complementary strategies include synthetic data augmentation using generative models to simulate under-represented pathologies and patient populations, multi-centre dataset consortia with harmonised annotation protocols, and transfer learning approaches designed to maximise performance in low-resource clinical settings.

D. Advancing Interpretability and Transparency

Primarily addresses: Reasoning stage

AI models frequently lack interpretability, leading to significant challenges concerning their performance and generalisability across diverse patient populations. These opaque technologies raise patient safety concerns, as non-interpretable models can result in improper treatment decisions due to misinterpretations by healthcare providers. For Reasoning-stage systems—which produce diagnostic suggestions or treatment recommendations—interpretability is not merely a desirable property but a regulatory and ethical requirement. Clinicians must be able to understand why a system produced a given output before acting on it, and regulatory bodies increasingly require that AI-generated clinical conclusions be traceable to specific evidence.

Research priorities in this area include post-hoc explanation methods that link textual outputs to specific image regions, causal attribution techniques that identify which features of an image or clinical record most influenced a model’s conclusion, and inherently interpretable architectures such as prototype-based networks that generate explanations as a native component of inference rather than an add-on. The integration of uncertainty quantification with interpretability models—enabling systems to express not only what they concluded but how confident they are and why—represents an underexplored area with significant clinical relevance.

E. Large-Scale Clinical Validation and Prospective Studies

Primarily addresses: Reasoning stage

The gap between benchmark performance and clinical reliability can only be closed through rigorous prospective evaluation in real healthcare environments. As noted in Section VIII, prospective multi-centre clinical trials for medical AI systems remain rare, and the existing literature is dominated by retrospective evaluations on curated datasets that do not capture the heterogeneity of clinical practice. Future progress requires a cultural and institutional shift toward treating clinical validation as a first-class research output rather than a post-publication consideration.

Specific research priorities include the design of randomised controlled trials evaluating the effect of AI-assisted diagnosis on patient outcomes—not only intermediate measures such as detection accuracy but downstream outcomes including time to treatment, misdiagnosis rates, and clinical pathway efficiency. Standardised reporting frameworks for medical AI clinical trials, analogous to CONSORT for randomised trials, are needed to enable meaningful comparison across studies. Finally, the development of living benchmarks—continuously updated evaluation datasets that reflect current clinical distributions—would provide a more dynamic and reliable measure of deployed model performance than static benchmark datasets that become progressively less representative as clinical practice evolves.

F. Summary

The five research directions outlined in this section collectively address the open challenges that currently prevent medical imaging AI from fulfilling its clinical potential. Progress at the Seeing stage—through more diverse training data and improved volumetric modelling—is a prerequisite for reliable performance at higher stages. Progress at the Saying stage—through reduced hallucination and stronger visual grounding—is necessary before AI-generated clinical language can be trusted. And progress at the Reasoning stage—through robust multimodal integration, interpretability, and prospective validation—will define whether medical AI can transition from a research tool to a reliable clinical partner. Achieving these goals will require sustained collaboration across the AI research community, the clinical community, healthcare institutions, and regulatory bodies.

XI. CONCLUSION

This paper has examined the evolution of artificial intelligence in medical imaging through the lens of the Seeing–Saying–Reasoning framework, which organises medical imaging AI systems according to the level of clinical intelligence they demonstrate rather than their architectural design. By conceptualising the field as a progression of functional mappings—from visual perception, through language-grounded communication, to knowledge-integrated clinical reasoning—the SSR framework provides a structured perspective on how the field is advancing from image interpretation toward clinically meaningful intelligence. Using this framework, we presented a capability-based taxonomy of representative multimodal systems, analysed the reliability and safety challenges that arise specifically at each stage, and examined the clinical translation and regulatory considerations that govern the path from research prototype to deployed clinical tool.

The analysis presented in this survey reveals several important patterns. Seeing-stage perception systems appear comparatively more mature, with architectures such as nnU-Net and Medical SAM showing strong performance across diverse imaging tasks. The transition to Saying-stage vision–language systems, however, introduces a qualitatively different set of challenges, including hallucination, weak visual grounding, and the inadequacy of standard evaluation metrics. Reasoning-stage systems remain at an earlier stage of development, with promising demonstrations from models such as Med-PaLM 2 and MAIRA-2 but without the prospective clinical validation required for trustworthy deployment. Bridging the gap between descriptive multimodal capability and clinically validated reasoning remains a central frontier for the field.

As with any survey, the present work has limitations that should be acknowledged. The literature reviewed spans the period from 2020 to 2025, and rapidly evolving areas—including large language model adaptation for clinical use and real-time AI-assisted surgical guidance—may not be fully represented. The selection of representative systems for the taxonomy, while guided by influence and methodological significance, inevitably reflects choices that another survey might make

differently. Furthermore, this survey does not include empirical experiments or meta-analyses of reported benchmark results, and the conclusions drawn regarding model capabilities and limitations are based on evidence reported in primary sources rather than independent verification. Future surveys in this area would benefit from more systematic selection criteria, broader modality coverage, and quantitative synthesis of benchmark performance across standardised evaluation protocols.

Looking forward, trustworthy medical imaging AI will require simultaneous progress across multiple dimensions—reducing hallucination and improving grounding at the Saying stage, advancing interpretability and multimodal fusion at the Reasoning stage, and conducting the large-scale prospective clinical trials necessary to validate AI-assisted diagnosis in real healthcare environments. The SSR framework introduced in this paper provides a principled structure for tracking this progress, identifying where the field stands at each level of clinical capability and where the most consequential gaps remain. As multimodal foundation models continue to advance, the central question is not only whether AI can perceive, describe, and reason about medical images, but whether it can do so reliably, transparently, and safely enough to support patient care.

REFERENCES